Barriers for measure, residue, corner and pairing methods in the lonely runner conjecture

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Abstract

We study which standard methods cannot prove the lonely runner inequality $\mathrm{ML}(v) \geq 1/(m+1)$ for positive integer speed vectors, and why. (i) At the target level, no integrable weight certifies a vector with $\mathrm{ML}(v) \leq 1/(m+1)$, so first-moment and Riesz-product certificates are blind to tight vectors; this is a formalized restatement of a lemma of Bedert. (ii) Five explicit families with 12 to 18 speeds defeat bounded-level base-point certificates of the kinds tested, and without a leave-one-out hypothesis no bound on denominators of lonely times depending only on m exists. (iii) For tight vectors of Goddyn–Wong type, $m+1$ divides no speed; a conjecture (T) extends this to all tight primitive vectors, and at a maximiser of a tight vector the denominator is $m+1$ times a divisor of the gcd of a corner pair. (iv) The doubling graph on the non-units in the window $[1, 2m]$ has an isolated vertex whenever $2m+1$ is composite and exceeds 9, hence no perfect matching. (v) Covering-radius bounds at the target value are false. (vi) An inequality $\min_v \mathrm{ML}(V \setminus v) \leq F(\mathrm{ML}(V))$ with F continuous at the floor cannot send counterexamples to tight subsets: at 11 values of m, exact tight vectors have all leave-one-out values above the threshold that F would have to respect (HEURISTIC, finite exact computation). The leave-one-out values of 70 known tight vectors are tabulated and a general lower bound is conjectured. (vii) The bad region has the first homology of the torus for $m \geq 2$ and its fundamental group for $m \geq 3$, so topological necessary conditions obtained from the inclusion are satisfied by every speed vector with not all speeds odd (the topological statements are PROVED; that this delimits a method is HEURISTIC). Each statement carries a label; attributions to prior work are listed in Section 11. No progress on the conjecture is claimed.

1 Introduction

1.1 Setting

A speed vector is $v = (v_1, \ldots, v_m) \in \mathbb{Z}^m_{>0}$; there are $m+1$ runners, one of them stationary. Write $\|x\| = \mathrm{dist}(x, \mathbb{Z})$, $\mathrm{gap}_v(t) = \min_i \|v_i t\|$ and $\mathrm{ML}(v) = \max_{t \in \mathbb{R}} \mathrm{gap}_v(t)$ (a maximum, since gap_v is continuous and 1-periodic). The lonely runner conjecture (LRC) states $\mathrm{ML}(v) \geq 1/(m+1)$ for all v. A vector is *tight* if $\mathrm{ML}(v) = 1/(m+1)$; $(1, \ldots, m)$ is tight. The bad set of speed i at level δ is $B_i = \{t : \|v_i t\| < \delta\}$ and the good set is $G_\delta = \{t : \forall i,\ \|v_i t\| \geq \delta\}$; a time t is *lonely at level* δ if $t \in G_\delta$, that is if $\mathrm{gap}_v(t) \geq \delta$. A vector is *divisor-complete* (DC) if every $j \in \{2, \ldots, m+1\}$ divides some v_i. Throughout, m is the number of speeds; the literature cited uses n for m or for $m+1$, and each citation below is translated.

The inequality is known for $m \leq 12$ (Rosenfeld for eight runners [13]; Trakulthongchai for nine and ten [16], refereed; Sungkawichai and Trakulthongchai up to thirteen runners [14],

unrefereed; Allikvere claims fourteen and fifteen runners [1], unrefereed, computer-assisted). The generic bounds are $1/(2m)$ by a union bound ([15, Prop. 1.1], [11, Section 5]) and $1/(2m) + m^{-5/3-o(1)}$ [3, Thm. 1.3].

The reduction studied in the accompanying archive reduces the conjecture for all m, given three finite statements, to the statement `FloorCruxU`: *if the floor holds for all smaller numbers of speeds, then for* $m \geq 12$ *every primitive* v *such that every leave-one-out subvector is primitive,* v *is DC,* v *admits a relation* $\sum c_i v_i = 0$ *with* $c \in \{-1, 0, 1\}^m \setminus \{0\}$*, and no Riesz-product certificate (Section 3) fires, satisfies* $\mathrm{ML}(v) \geq 1/(m+1)$ (Lean: `FloorCruxU`, `universalFloor_of_four_atoms_U`). This statement is open. The methods below are those that were aimed at it.

1.2 Labels and provenance

VERIFIED: formalized in Lean 4 with Mathlib, zero `sorry`, axioms among `propext`, `Classical.choice`, `Quot.sound`, in the repository making the claim. All VERIFIED statements in this note are *carried over from the repository where they built; not rebuilt in this archive*: the build artefacts of the modules concerned are present there with timestamps matching their sources and contain no `sorry` (the modules holding the open statements (T), (T′), (T2) do contain `sorry`); the axiom prints of 1 October 2026 are recorded in `LEAN_MAP.md` and were not re-run. PROVED: complete written proof, reviewed adversarially, with conceptual abstract and zoom map, not formalized. HEURISTIC: computation or plausibility only; searches state their ground and termination status (RESOLVED if completed, UNRESOLVED if capped). CONJECTURE: no proof. FALSE: counterexample in hand. The strength of a result is the weakest label in its dependency chain.

1.3 Results

Item	Section	Statement	Label
Thm 2	3	If $\mathrm{ML}(v) \le \delta$ the good set is countable; no integrable weight can satisfy the certificate inequality at a tight vector	VERIFIED, carried over
Cor 3	3	No Riesz-product certificate fires on tight vectors	VERIFIED, carried over
Rem 4	3	Scope: what the barrier does not cover	HEURISTIC
Prop 5, 6	4	Escape lemma; missing divisor j gives $\|v_i/j\| \ge 1/j$ and a lonely time	VERIFIED, carried over (escape; weak form of the missing-divisor bound); strict form PROVED
Prop 7	4	(a) Times of positive gap of $v^{(Q)}$ have denominator $> Q$; (b) $\mathrm{ML}(v^{(Q)}) \ge 1/(m+1)$; so no bound $Q(m)$ on denominators of lonely times without leave-one-out primitivity	PROVED
Obs 8	4	Five families defeat bounded-level certificates	HEURISTIC (exact, finite)
Thm 9, Cor 10	5	Goddyn–Wong type vectors: $m+1$ divides no speed	VERIFIED, carried over
Prop 13	5	The implications (T′) ⇒ (T) and (T) ⇒ no tight primitive vector is DC; neither (T) nor (T′) is proved	PROVED (implications)
Prop 14	5	At a maximiser k/L of v with $\mathrm{ML}(v) = a/b < \frac{1}{2}$: $b \mid L$ and L/b divides the gcd of a corner pair	PROVED
Conj 12, 15	5	(T), (T′), (T2)	CONJECTURE
Prop 16	6	Isolated vertex in the doubling graph on the non-units of $[1, 2m]$ for composite $2m+1 > 9$	PROVED
Prop 18, Rem 19	7	Half-phase statement false; covering-radius target false	VERIFIED, carried over; cited computation
Lem 20, Rem 21	8	Deleting a speed does not lower ML; $\mathrm{ML}(cV) = \mathrm{ML}(V)$; no consequence for subsets of counterexamples	PROVED
Lem 22	8	A descent function F at m has $F(1/(m+1)) \ge \mathrm{loo}(V)$ for tight V	PROVED
Obs 23, Prop 24	8	At $m \in \{7, 31, 61, 73, 91, 121, 143, 151, 181, 211, 241\}$ a sub-floor descent function jumps at the floor	HEURISTIC (deduction PROVED; instances exact, not formalized)
Obs 25	8	Leave-one-out values of 70 known tight vectors: 11 tight, 46 at $2/(2m-1)$, 13 above, 0 below	HEURISTIC
Conj 26	8	Tight primitive vectors without a tight leave-one-out have $\mathrm{loo} \ge 2/(2m-1)$	CONJECTURE
Obs 27	8	$\mathrm{ML}(V) < 1/m$ does not imply a tight leave-one-out ($m = 5, 8, 10$)	FALSE (statement), exact witnesses
Prop 29, Cor 30, Prop 31	8.4	$U_\delta \to \mathbb{T}^m$ is an isomorphism on H_1 ($m \ge 2$) and π_1 ($m \ge 3$); an equivariant loop is homotopic into U_δ if and only if not all v_i are odd	PROVED
Rem 28, 32	8	Scope of the two obstructions	HEURISTIC

Theorem 2 and the escape lemma are known in substance (Tao, Bedert, Rosenfeld); Proposition 6 is known; Theorem 9 is, to the extent read, not stated in the literature, but its proof is a few lines. What is attributed and what is not is itemized in Section 11.

1.4 Conceptual abstract

The note tests seven standard routes to the floor at tight vectors. A fixed integrable weight cannot certify a tight vector: its good set is countable and without mass. Bounded-denominator certificates fail without a leave-one-out hypothesis, because a speed divisible by $\mathrm{lcm}(1,\dots,Q)$ vanishes at every time of denominator at most Q. Doubling pairings fail because, for odd composite $2m+1$ above 9, the non-unit $2m+1-2p$, with p the least prime factor, has no partner in the window. Corner analysis fails because a corner maximum depends on the residues of all other speeds. Covering-radius bounds fail because the shifted statement is false. Descent through leave-one-out subsets fails because some tight vectors have every leave-one-out value above the next value attained by $m-1$ speeds. Topological conditions fail because the bad region has the first homology of the torus. What remains is an explicit rational time or a classification of tight vectors.

1.5 Zoom map for Theorem 2 and Corollary 3

L1.1 A level set of $\|at\|$ is countable (`nearestIntDist_eq_mem`).

L1.2 If $\mathrm{ML}(v) \le \delta$, then G_δ lies in the union of such level sets, hence is countable and null (`goodSet_countable`, `goodSet_null`).

L1.3 A nonnegative weight satisfies $\int F \le \sum_i \int_{B_i} F$ (`integral_le_sum_badSet`).

L1.4 Contrapositive: a strict inequality $\sum_i \int_{B_i} F < \int F$ implies $\delta < \mathrm{ML}(v)$ (`lt_ML_of_integral_unionBound`).

L1.5 Specialization to Riesz-product weights at $\delta = 1/(m+1)$ and non-vacuity on $(1,\dots,m)$ (`not_UDFires_of_tight`, `ML_consec_le`).

2 Methods against known theorems

Method	Strongest known statement it supports	Where it stops	Source
First moment on individual bad sets	$\mathrm{ML} \geq 1/(2m)$	ignores overlaps of bad sets; insensitive to phases, so cannot exceed the shifted infimum (Section 7)	[15], [11]
Weighted first moment, Riesz products	$\mathrm{ML} \geq 1/(2m) + m^{-5/3-o(1)}$; floor for dissociated sets	Thm 2: fixed integrable weights see no room at tight vectors	[3]
Atomic measures, rational witness times	floor when some $j \leq m+1$ divides no speed; floor for $m \leq 12$ by sieved computation	Prop 7, Obs 8: fixed-level checks are defeated	[15], [13], [16], [14]
Relation structure (Fourier on $\mathbb{Z}/q$, windows)	counterexample or tight $\Rightarrow$ integer relation of ℓ^1 norm $\leq 2m+3$	tight vectors near $(1, \ldots, m)$ satisfy it vacuously	[6], [2]
Finite checking, lattice width	counterexamples have bounded height; $n^{O(n^2)}$, $(n(n+1)/2)^{n-1}$	bounds not usable for all m	[15], [10], [7]
Covering radius	equals shifted loneliness (Henze–Malikiosis)	the shifted statement is false (Sec. 7)	[4]
Local analysis, Goddyn–Wong families	tight vectors of one-entry type are classified	transfers tightness, not a floor (Sec. 5)	[8], [17]
Doubling pairings	(internal to the covering framework)	isolated vertices (Sec. 6)	this note

3 Volume and smooth-weight certificates

For a nonnegative $F \in L^1[0,1]$ and a level δ consider

$$\sum_i \int_{[0,1]\cap B_i} F < \int_{[0,1]} F. \tag{1}$$

If $\mathrm{ML}(v) < \delta$, every t lies in some B_i, so $F \leq \sum_i \mathbf{1}_{B_i} F$ pointwise and (1) fails. The weights $F = P^D$, $P(t) = \prod_i (1 - \cos 2\pi v_i t)$, are Riesz products; U_D *fires* if (1) holds for $F = P^D$ at $\delta = 1/(m+1)$. The case $F \equiv 1$ is the union bound of [15, Prop. 1.1]. For an arbitrary probability measure μ and closed bad sets the inequality $\int \Phi \, d\mu \geq 1$, $\Phi = \sum_i \mathbf{1}_{\{\|v_i x\| \leq \delta\}}$, at $\delta = \mathrm{ML}(v)$ is [3, Lemma 2.1].

Lemma 1 (Level sets)**.** *For $a \in \mathbb{Z} \setminus \{0\}$ and $\delta \in \mathbb{R}$, $\{t : \|at\| = \delta\} \subset \{(n \pm \delta)/a : n \in \mathbb{Z}\}$ is countable.*

Theorem 2. *Let $v \in (\mathbb{Z} \setminus \{0\})^m$, $\delta \in \mathbb{R}$ and $\mathrm{ML}(v) \leq \delta$. Then G_δ is countable, hence Lebesgue-null. Consequently, for every measurable $F \geq 0$ integrable on $[0,1]$,*

$$\int_{[0,1]} F \leq \sum_i \int_{[0,1]\cap B_i} F,$$

so (1) *implies $\delta < \mathrm{ML}(v)$.*

Proof. Let $t \in G_\delta$. Then $\mathrm{gap}_v(t) \geq \delta \geq \mathrm{ML}(v) \geq \mathrm{gap}_v(t)$, so $\mathrm{gap}_v(t) = \delta$. As gap_v is a minimum over finitely many terms, some i has $\|v_i t\| = \delta$. By Lemma 1, G_δ is a finite union of countable sets. Outside G_δ some i has $t \in B_i$, hence $F(t) \leq \sum_i \mathbf{1}_{B_i}(t) F(t)$ for almost every t; integrate over $[0,1]$. □

Label: VERIFIED, carried over (`goodSet_countable`, `goodSet_null`, `integral_le_sum_badSet`, `lt_ML_of_integral_unionBound`). The inequality for absolutely continuous measures is Bedert's Lemma 2.1 read with $\mu = F\,dt$; the countability of G_δ is elementary.

Corollary 3. *If* $\mathrm{ML}(v) \leq 1/(m+1)$*, in particular if* v *is tight, no* U_D *fires,* $D \geq 0$*. Conversely, if* U_D *fires then* $1/(m+1) < \mathrm{ML}(v)$*.*

Label: VERIFIED, carried over (`not_UDFires_of_tight`, `lt_ML_of_UDFires`); the hypothesis is not vacuous, $\mathrm{ML}(1,\dots,m) \leq 1/(m+1)$ *(`ML_consec_le`, `not_UDFires_consec`). The corollary is an immediate consequence of the cited lemma of Bedert.*

Remark 4 (Scope). Theorem 2 concerns (1) at one fixed level and one fixed weight. It does not exclude a weight $F_{\delta'}$ chosen separately for each level $\delta' < \mathrm{ML}(v)$: the indicator of $\{\mathrm{gap}_v > \delta'\}$ satisfies (1) exactly for $\delta' < \mathrm{ML}(v)$, but constructing it requires knowing the lonely times. What is excluded is a weight given by a formula uniform in v, such as P^D, at the target level. Atomic measures are outside the theorem: a point mass at a rational time is the escape lemma (Proposition 5), and the uniform measure on $\{j/p\}$ with p a prime dividing no speed is [3, Lemma 5.3, Cor. 5.4]. Volume comparisons of geometry-of-numbers type (successive minima, covering radii bounded through volume) were not shown to reduce to (1); the claim that they do is not made. Label: HEURISTIC for the scope statements, as they delimit methods and are not theorems.

Obstruction for volume methods. A first-moment argument that uses only the measures of the individual bad sets gives $1/(2m)$, because each B_i has measure 2δ and the sum must reach 1; the factor two is the multiplicity with which bad sets overlap, which this argument discards. Sharper volume arguments must control the intersections $B_i \cap B_j$, and these depend on the starting phases of the runners, whereas the individual measures do not; since the shifted statement is false (Section 7), a volume argument can pass the exponent only by using the zero phase through the intersection pattern. At the target value itself the pattern is rigid: by Theorem 2 the bad sets of a tight vector cover the circle up to a null set, so a weight can gain nothing beyond equality.

Obstruction for smooth-weight certificates. A Riesz product or any other integrable weight turns the existence of a lonely time into a strict inequality between two Lebesgue integrals, and at a vector with $\mathrm{ML}(v) = 1/(m+1)$ the two integrals are forced by Theorem 2 to satisfy the opposite non-strict inequality, because the lonely set is countable and carries no mass. The mechanism is that gap_v is a minimum of finitely many tent functions, each of whose level sets is discrete; no choice of weight can see a set of measure zero. The obstruction persists off the tight set in the tested range: the five families of Table 1 have ML strictly above the target and $U_1, U_2 \geq 1$.

4 Residue and base-point certificates

Proposition 5 (Escape lemma)**.** *Let $q \geq 1$, $a \in \mathbb{Z}$, $\varepsilon \in \mathbb{R}$, $\delta > 0$. If for each i there are integers N_i, r_i with $v_i a = qN_i + r_i$ and $\delta \leq r_i/q + v_i\varepsilon \leq 1 - \delta$, then $\delta \leq \mathrm{ML}(v)$.*

Proof. At $t = a/q + \varepsilon$ we have $v_i t = N_i + r_i/q + v_i\varepsilon$, so $\|v_i t\| \geq \delta$ for all i. □

Label: VERIFIED, carried over (`ML_ge_of_escape`). With $\varepsilon = 0$ this is the definition of a proper tuple in [16, Def. 3], the witness-time ansatz of [13]; the extension to $\varepsilon \neq 0$ is a routine generalization.

Without a bound on q the statement is empty: $\delta \leq \mathrm{ML}(v)$ holds if and only if some (q, a, ε) works, with $q = 1$ (`lonely_iff_escape`, VERIFIED, carried over), and for $\delta < \mathrm{ML}(v)$ one can take $\varepsilon = 0$ (`rational_point_of_lt`, VERIFIED, carried over). The window lemma at levels $m+1 < q \leq 2m+2$ and the certificates used in the archive are instances (`window_via_escape`, `famF_via_escape`, `lemmaP_via_escape`). All content lies in the bound on q.

Proposition 6 (Missing divisor)**.** *If $2 \leq j \leq m+1$ and j divides no v_i, then $\|v_i/j\| \geq 1/j$ for all i, so $\mathrm{ML}(v) \geq 1/j$ with witness $t = 1/j$. In particular $\mathrm{ML}(v) > 1/(m+1)$ if $j \leq m$, and $\mathrm{ML}(v) \geq 1/(m+1)$ if $j = m+1$.*

Label: The bound $\|v_i/j\| \geq 1/j$ is PROVED in one line: $v_i \bmod j \in [1, j-1]$. The Lean lemma `witness_of_missing_divisor` (VERIFIED, carried over) has the signature $j \leq (k+2)\min(r, j-r)$ with $r = v_i \bmod j$ for $k+1 = m$ speeds, that is $\|v_i/j\| \geq 1/(m+1)$, the weak form; the strict form is not stated there, and no other part of that Lean statement was re-read in this revision. Known: [15, Lemma 5.1(i)]; also the survey [11] proves that the conjecture holds when some $k \leq n+1$ divides no speed, and [17, Lemma 2.3] is the case $t = 1/r$ for one-entry modifications. Not claimed as new.

Consequently a tight vector satisfies: every $j \in [2, m]$ divides some speed, since otherwise $\mathrm{ML}(v) \geq 1/j > 1/(m+1)$. This is why the crux is restricted to DC vectors.

Proposition 7 (Denominators of lonely times)**.** *Let $m \geq 2$, $Q \geq 1$, $L = \mathrm{lcm}(1, \ldots, Q)$ and $v^{(Q)} = (1, L, 2L, \ldots, (m-1)L) \in \mathbb{Z}^m_{>0}$. If $t = a/q \in \mathbb{Q}$ in lowest terms has $\mathrm{gap}_{v^{(Q)}}(t) > 0$, then $q > Q$. (b) $\mathrm{ML}(v^{(Q)}) \geq 1/(m+1)$, witnessed by $t = c/(L(m+1))$ for any integer c with $L \leq c \leq Lm$ and $c \equiv 1 \pmod{m+1}$; such c exists. Consequently, for every function $Q(m)$ there is a vector with m speeds that has lonely times at level $1/(m+1)$ but none of denominator at most $Q(m)$.*

Proof. If $q \leq Q$ then $q \mid L$, so $Lt \in \mathbb{Z}$ and $\|Lt\| = 0$. The speed L occurs in $v^{(Q)}$ because $m \geq 2$, so $\mathrm{gap}(t) = 0$. Hence $q > Q$ whenever $\mathrm{gap}(t) > 0$.

(b) The interval $[L, Lm]$ contains $L(m-1)+1$ integers. For $L = 1$ it contains $1 \equiv 1$. For $L \geq 2$ it contains at least $m+1$ consecutive integers, hence one with $c \equiv 1 \pmod{m+1}$. Put $t = c/(L(m+1))$. Since $L \leq c \leq Lm$, $t \in [\frac{1}{m+1}, \frac{m}{m+1}]$ and $\|t\| \geq \frac{1}{m+1}$ (speed 1). For $1 \leq j \leq m-1$ the speed jL gives $jLt = jc/(m+1)$; $\gcd(c, m+1) = 1$ and $m+1 \nmid j$, so $m+1 \nmid jc$ and $\|jc/(m+1)\| \geq \frac{1}{m+1}$. Hence $\mathrm{gap}(t) \geq \frac{1}{m+1}$, and by (a) its denominator is $> Q$. For the consequence take $Q = Q(m)$. □

Label: PROVED. Idea: a speed divisible by every small denominator is invisible to every small-denominator test, because it sends all rational times of small denominator to 0 mod 1. Zoom

map: L1.1 reduction of $q \le Q$ to $q \mid L$ (arithmetic, no lemma); L1.2 positivity of the gap forces $q > Q$; L1.3 existence of c and the bound $\mathrm{gap}(t) \ge \frac{1}{m+1}$ at $t = c/(L(m+1))$. For $Q \ge 2$ the subvector $(L, 2L, \ldots, (m-1)L)$ obtained by deleting the speed 1 has $\gcd = L > 1$, so $v^{(Q)}$ is outside the class of the crux, whose members have primitive leave-one-out subvectors. No prior statement was located in the literature read.

Rationality of maximizers. The gap is piecewise linear with slopes $\pm v_i \neq 0$, so it has no flat piece and each local maximum is a vertex of its graph; every vertex is the solution of a linear equation with integer coefficients, hence rational. So every maximiser of gap_v is rational, and for $v = v^{(Q)}$ every maximiser, which has positive gap, has denominator $> Q$ by the proposition. This is not needed for the proposition.

Obstruction for residue-condition methods. A test at denominator q sees only the residues $v_i \bmod q$ of the speeds, and a speed with residue 0 at every level up to Q is invisible to all of them; Proposition 7 makes this exact, and the families of Table 1 realize it approximately inside the class of the crux by a few tall speeds (up to 4943) whose residues complete a covering of the small levels. The mechanism is that the residues of a few free tall speeds can be chosen independently modulo each small q while the maximal loneliness stays above the target; this explanation is consistent with the table and is not proved, label HEURISTIC.

4.1 Level drift

Let $\mathrm{TL}(Q)$ denote the statement: for $m \ge 12$ and every class member v of the crux, there are $q \le Q(m)$, a unit a modulo q, $\sigma = \pm 1$ and $\varepsilon \ge 0$ such that Proposition 5 applies at $t = \sigma a/q + \varepsilon$ with the first-order arc condition on the non-multiples of q and a condition on the multiples of q through ε. The certificate kinds named in the table (WC, window points, window levels, exact multiple scan, first order) are the variants implemented in `cert_m` of `code/reduction/mdr`.

Observation 8. For each certificate kind and level range in Table 1, the stated class member has no certificate of that kind in that range.

Label: HEURISTIC. Each row is a finite exact computation (exact integer arithmetic; ML *recomputed by two independent programs, see Section 9), not checked in Lean. Class membership used U_1, U_2 computed in double precision with an a posteriori bound of 10^{-12}; the listed margins exceed it. Termination: rows 1 to 5 RESOLVED (exhaustive over the stated levels); row 6 UNRESOLVED.*

Last row of the table: a time-limited hunt of 120 s per m (one process, seed 1) performed 14 686, 8 962, 4 141, 2 859 evaluations for $m = 18, 20, 22, 24$, reached best levels $29, 37, 34, 37$ against the cutoffs $38, 42, 46, 50$, and produced no family; an earlier run of 480 s per m was reported with the same outcome and is not repeated here. Both are UNRESOLVED and support no statement about other budgets or seeds.

Positive ground, regenerated by `mdr enumw`: the window-only first-order version had no failure on the exhaustive supersets (DC, leave-one-out primitive, not strongly free) for $m = 6$, $H \le 50$ (15 890 700 sets, 3 202 695 in the class); $m = 8$, $H \le 24$ (735 471; 119 915); $m = 10$, $H \le 30$ (30 045 015; 3 327 077); $m = 12$, $H \le 26$ (9 657 700; 934 023); $m = 14$, $H \le 24$ (1 961 256; 107 976), each box RESOLVED; and none on the 1067 census families (lists in `data/obstructions/census`) and the 147 adversarial families (`mdr cert`, certificate level at

Certificate kind and range	Class member without certificate	U_1, U_2
WC, $q \le m+1$	$[1,4,5,6,7,9,11,13,40,60,238,2731]$, $m=12$, ML $=14/121$	1.003918, 1.001345
Window points, $q \le 2m+2$	$[1,3,4,5,7,9,11,13,40,48,100,1079]$, $m=12$, ML $=11/84$ (first certified at $q=13$ by WC)	1.002573, 1.000429
Window levels, exact, $q \le 2m+2$	$[1,5,6,7,8,9,11,13,17,19,23,25,29,31,366,1680]$, $m=16$, ML $=168/1697$; first level 35	1.483, 1.715
All levels, first order and exact, $q \le 2m+1$	$[1,3,4,5,7,11,13,17,19,23,31,70,74,116,180,272,3986,4943]$, $m=18$; ML $=28/191$; first point at $q=38$	1.007002, 1.003326
All levels, first order, $q \le 2m+2$	$[1,3,4,5,7,11,13,17,19,23,31,38,70,74,116,180,272,4422]$, $m=18$; ML $=30/203$; first level 39	1.024583, 1.014631
Exact multiple scan, $q \le 2m+2$, $m=18,\dots,24$	none found in time-limited hunts; UNRESOLVED (not exhaustive)	n/a

Table 1: Families defeating bounded-level certificates (script `code/obstructions/repro_04.sh`). The window-levels row also has a first-order version, first level 41, on $[1,3,4,5,7,8,9,11,12,13,17,19,23,29,31,37,224,1800]$, $m=18$. The $m=12$ vector in row 2 has least witness $(4,31)$.

most $2m+2$ for all). The census families are residuals of boxes and are not independent of the boxes. The failures above thus appear only at larger m or height. The role of the condition "no U_D fires" in this ground is not understood.

5 Local corners: the Goddyn–Wong mechanism and statement (T)

Corners. $\mathrm{ML}(v)$ is attained at a time k/D with $D \in \{2v_i, v_i+v_j, |v_i-v_j|\}$ and $0 \le k < D$ ([11, Section 2]; the numerators are not restricted to residues coprime to D). A local argument fixes such a pair and controls the maximum from it; the remaining $m-2$ speeds enter only through their residues modulo D. For $\mathrm{ML}(v) < \frac{1}{2}$ every maximizer is such a corner, of a rising and a falling tent (Proposition 14).

For multipliers $\mu = (\mu_1,\dots,\mu_{n-1}) \in \mathbb{N}^{n-1}$ write $[n-1]_\mu = (\mu_1\cdot 1,\dots,\mu_{n-1}(n-1))$, with $n = m+1$. The *GCD condition* is: for each $r \in [n-1]$ with $\mu_r \ge 2$ and each b with $n-r \le b \le \mu_r(n-r)-1$, $\gcd(r,b) > 1$. Goddyn and Wong prove $\mathrm{ML}([n-1]_\mu) = 1/n$ under this condition [8, Thm. 3.1], quoted as [17, Thm. 5.1]; for one accelerated entry it is also necessary [8, Thm. 2.3], quoted as [17, Thm. 1.1]. Zhang classifies all one-entry modifications $[n-1]_{r\to w}$, $w > n-1$: tight if and only if $2r > n-1$, $r \mid w$ and the GCD condition holds with $\mu = w/r$, or $(n,r,w) \in \{(5,2,7),(6,2,9)\}$ [17, Thm. 1.2].

Theorem 9. *If μ satisfies the GCD condition, then n divides no speed $\mu_r r$, $1 \le r \le n-1$.*

Proof. If $\mu_r = 1$ then $0 < r < n$. Let $\mu_r \ge 2$ and suppose $\mu_r(n-r) \ge n$. The interval $[n-r, \mu_r(n-r)-1]$ has length $\mu_r(n-r)-(n-r) \ge r$, so it contains r consecutive integers, one of which is $\equiv 1 \pmod r$ and hence coprime to r; this contradicts the GCD condition. So $0 < \mu_r(n-r) < n$. Now $\mu_r r + \mu_r(n-r) = \mu_r n$, so $n \mid \mu_r r$ would give $n \mid \mu_r(n-r)$, impossible for $0 < \mu_r(n-r) < n$. □

Label: VERIFIED, carried over (`not_dvd_speed`; axioms recorded as `propext`, `Quot.sound`). A statement of this form was not located in the sources read; the original proof of [8, Thm. 3.1]

was not available for this audit and may contain the same step. The published argument obtains a sharper size bound; only the weaker one is used.

Corollary 10. *Under the GCD condition with $\mu_1 = 1$ (forced, since $\gcd(1, b) = 1$), the vector $[n-1]_\mu$ is not DC: $j = n = m+1$ divides no speed. The same holds for the vectors $(1, 3, 4, 7)$, $(1, 3, 4, 5, 9)$, $(1, 4, 5, 6, 7, 11, 13)$, $(1, 2, 3, 4, 5, 7, 12)$, $[1..11] \cup \{13, 24\}$, $[1..17] \cup \{19, 36\}$ and for the two-entry doubling instance $[1..73] \setminus \{70, 72\} \cup \{140, 144\}$ with $n = 74$ [17, Cor. 5.2 and the example following Thm. 5.1].*

Label: VERIFIED, carried over for $[n-1]_\mu$ under the GCD condition (`not_divisorComplete`). For the listed vectors: HEURISTIC, an exact computation (script `code/obstructions/gwcheck.py`) of $\mathrm{ML} = 1/n$ and of the non-divisibility, RESOLVED.

Remark 11. A vector built by the Goddyn–Wong construction is excluded from the class of the crux, since the proof of its tightness shows $m+1 \nmid v_i$. The mechanism uses three inputs: the baseline $[n-1]$, the identity $\mu_r r \equiv -\mu_r(n-r) \pmod{n}$ valid because the new speed is an integer multiple of an existing entry, and the GCD hypothesis reused through a size bound. A general class member has no such decomposition, and the generalizations in the literature (Zhang's one-entry classification, the multi-swap theorem) keep the multiplicative form. The route therefore gives no lower bound for divisor-complete vectors. The converse of the multi-acceleration theorem is not true in general [17, Question 9.3].

Obstruction for local-corner methods. The maximum of gap_v is a corner determined by two speeds, but whether the corner is a maximum depends on the residues of every other speed modulo the corner's denominator, and these residues carry no structure that a pair-local argument can use. In the Goddyn–Wong families the missing control is supplied by a global identity, namely that every speed is a multiple of an entry of $[n-1]$ and the sum $\mu_r r + \mu_r(n-r)$ is a multiple of n; for a vector without this decomposition the maximizer still sits at denominator $m+1$ in every tight case computed (Section 9), but no pair of speeds identifies which speed would carry the divisor $m+1$ if it existed. This is why a direct local proof of (T) below gets no purchase, label HEURISTIC.

5.1 Statement (T)

Conjecture 12 ((T), (T′))**.** *Let $v \in \mathbb{Z}^m_{>0}$ with $\gcd(v) = 1$ and $\mathrm{ML}(v) = 1/(m+1)$. Then (T) $m+1$ divides no v_i; (T′) every t with $\mathrm{gap}_v(t) = \mathrm{ML}(v)$ has the form $a/(m+1)$, $a \in \mathbb{Z}$.*

Label: CONJECTURE. Provenance: observation on the tight vectors of Goddyn–Wong type and on exhaustive small boxes. Evidence: Section 9, exhaustive primitive search for $m = 3, \ldots, 8$ in boxes finds the tight vectors listed there, all satisfying (T) and (T′). Consequence: by Proposition 6 a tight vector has every $j \in [2, m]$ dividing a speed, so (T) is equivalent to the statement that no tight primitive vector is DC. Lean statements `TightNoDvd`, `TightMaximisersAtMp1` are defined without proof; `tightNoDvd_of_strong` has a `sorry`.

Proposition 13. *Let $v \in \mathbb{Z}^m_{>0}$ and suppose $m+1 \mid v_i$ for some i. Then $\mathrm{gap}_v(a/(m+1)) = 0 < \mathrm{ML}(v)$ for every $a \in \mathbb{Z}$. Consequently (T′) implies (T), and (T) implies that no tight primitive vector is DC.*

Proof. Since $v_i/(m+1) \in \mathbb{Z}$, $v_i \cdot a/(m+1) \in \mathbb{Z}$ and $\mathrm{gap}_v(a/(m+1)) \le \|v_i t\| = 0$. Also $\mathrm{ML}(v) > 0$, as $\mathrm{gap}_v(t) > 0$ at any irrational t. Assume (T′) and let v be primitive and tight with $m+1 \mid v_i$. A maximizer t exists, since gap_v is continuous and 1-periodic; by (T′), $t = a/(m+1)$, so $\mathrm{gap}_v(t) = 0 \neq \mathrm{ML}(v)$, a contradiction. Hence (T) holds. A DC vector has $m+1 \in \{2, \dots, m+1\}$ dividing some v_i, which (T) excludes for tight primitive v. □

Label: PROVED (the implications only; neither (T) nor (T′) is proved). Idea: a speed divisible by $m+1$ vanishes at every candidate time of denominator $m+1$, so such a time cannot be a maximizer. Zoom map: L1.1 a divisible speed has gap 0 at $a/(m+1)$; L1.2 a maximizer exists and has positive gap; L1.3 divisor-completeness supplies the divisible speed. The proof does not use $\gcd(v) = 1$; that hypothesis matters for the truth of (T′), since the vector (2), $m = 1$, is tight and non-primitive and its maximizer $1/4$ is not in $\frac{1}{2}\mathbb{Z}$. Not formalized.

Proposition 14 (Tight corner)**.** *Let $v \in \mathbb{Z}^m_{>0}$ with $\mathrm{ML}(v) = a/b < \frac{1}{2}$ in lowest terms, and let t^* be a maximizer of gap_v. Then $t^* = k/L$ in lowest terms is rational, and there are indices i, j with $v_i t^* \equiv a/b$ and $v_j t^* \equiv -a/b \pmod 1$ such that*

$$b \mid L, \qquad L \mid b \gcd(v_i, v_j), \qquad L \mid v_i + v_j.$$

In particular $L = bd$ with $d \mid \gcd(v_i, v_j)$. If v is tight and $m \ge 2$ then $b = m+1$, so (T′) holds at t^* if and only if $d = 1$.

Proof. L1: (a) at a maximizer some speed sits at $+a/b$ and some at $-a/b$; (b) the two congruences give the divisibilities.

(a) Put $\delta = a/b$ and $S = \{i : \|v_i t^*\| = \delta\}$, nonempty since $\mathrm{gap}_v(t^*) = \delta$. As $0 < \delta < \frac{1}{2}$, each $i \in S$ has $v_i t^* \equiv \sigma_i \delta \pmod 1$ with a unique $\sigma_i \in \{+1, -1\}$. Suppose $\sigma_i = +1$ for all $i \in S$. For small $\varepsilon > 0$, $\|v_i(t^* + \varepsilon)\| = \delta + v_i \varepsilon$ for $i \in S$, and $\|v_i(t^* + \varepsilon)\| > \delta$ for $i \notin S$ by continuity. Then $\mathrm{gap}_v(t^* + \varepsilon) > \delta = \mathrm{ML}(v)$, which is absurd. The case $\sigma_i = -1$ for all $i \in S$ is the same with $t^* - \varepsilon$. So there are $i, j \in S$ with $\sigma_i = +1$, $\sigma_j = -1$. Then $v_i t^* = N_i + a/b$ and $v_j t^* = N_j - a/b$ with $N_i, N_j \in \mathbb{Z}$; in particular $t^* \in \mathbb{Q}$, say $t^* = k/L$ with $\gcd(k, L) = 1$.

(b) Adding, $(v_i + v_j)k/L = N_i + N_j \in \mathbb{Z}$, so $L \mid (v_i + v_j)k$ and $L \mid v_i + v_j$. Multiplying $v_i k/L = N_i + a/b$ by bL gives $bv_i k = L(bN_i + a)$. Hence $L \mid bv_i k$, so $L \mid bv_i$; and reducing modulo b gives $b \mid La$, so $b \mid L$ because $\gcd(a, b) = 1$. In the same way $bv_j k = L(bN_j - a)$ gives $L \mid bv_j$. Thus $L \mid \gcd(bv_i, bv_j) = b \gcd(v_i, v_j)$, and $L = bd$ gives $d \mid \gcd(v_i, v_j)$.

For tight v and $m \ge 2$, $\mathrm{ML}(v) = 1/(m+1) \le \frac{1}{3} < \frac{1}{2}$, so $a = 1$, $b = m+1$. (T′) at t^* says $L \mid m+1$; with $m+1 \mid L$ this is $L = m+1$, that is $d = 1$. □

Label: PROVED. Idea: at a maximizer one runner sits exactly at distance $+$ML ahead of an integer and another exactly ML *behind (the two residues $\pm$ML are distinct because $\mathrm{ML} < \frac{1}{2}$), otherwise the time could be moved to raise the gap; clearing the denominator of* ML *turns the two corner equations into congruences that bound the denominator of the maximizer by b times a common divisor of the corner pair. Zoom map: L1.1 opposite signs at a maximizer; L1.2 divisibilities from the two corner equations. That a maximizer lies at a time k/D with D a sum of two speeds is the vertex lemma [5, Lemma 2.2], [11, Section 2]; the divisibilities $b \mid L$ and $L \mid b \gcd(v_i, v_j)$ were not located in the sources read. For tight v with $m+1 \mid v_l$ and $d = 1$ at a maximizer, $v_l t^* \in \mathbb{Z}$, which is impossible; so a tight primitive vector with a speed divisible by $m+1$ would have $d \ge 2$ at every maximizer. Worked example: $v = (1, 2, 4, 6)$, $\mathrm{ML} = 1/4$, $t^* = 3/8$, $L = 8$, $b = 4$, $d = 2$, corner pair $(6, 2)$ with $6t^* \equiv \frac{1}{4}$, $2t^* \equiv -\frac{1}{4}$, $L \mid v_i + v_j = 8$,*

$L \mid b\gcd(6,2) = 8$ *(not tight:* ML $= 1/4 > 1/5$*). Tight non-primitive example:* $v = (2,4)$*,* $m = 2$*,* ML $= 1/3$*, with maximizers* $1/6, 1/3, 2/3, 5/6$*; at* $t^* = 1/6$ *the corner pair is* $(2,4)$*,* $L = 6$*,* $b = 3$*,* $d = 2$*. So* $d > 1$ *occurs, and the hypothesis* $\gcd(v) = 1$ *is needed for* (T')*. Not formalized.*

5.2 The family E_m and a related floor

For $E_m = (1, \ldots, m-1, m(m+1))$, $m \geq 3$, $\mathrm{ML}(E_m) = (m+1)/(m^2+m+1)$ (the case $s = m+1$ of [9, Thm. 5.1], reproved in Lean with a different argument, `ML_Em_m`, VERIFIED, carried over). Hence $(2m+1)\mathrm{ML}(E_m) > 2$ for $m \geq 3$ (`thmB_gt`, VERIFIED, carried over).

Conjecture 15 ((T2))**.** *If* v *is primitive and* $m+1$ *divides some* v_i*, then* $\mathrm{ML}(v) \geq 2/(2m+1)$*.*

Label: CONJECTURE. Evidence: in the exhaustive boxes of Section 9, the minimum of ML *over primitive DC vectors equals* $2/(2m+1)$ *for* $m = 3, \ldots, 7$*, attained for example by* $(1,3,4)$*, and* E_m *is not a minimizer. Consequence: every primitive vector in which* $m+1$ *divides a speed satisfies the floor with strict inequality. The vector* $(1,5,6,11,16,17)$*,* $m = 6$*, has* $\mathrm{ML} = 5/33 < 2/13$*, but no speed is divisible by* 7*, so it lies outside the class. A statement that* E_m *is the unique minimizer is false.*

6 Doubling pairings

In the covering formulation a lonely time is ruled out by exhibiting bad sets that cover the window of integers $[1, 2m]$ for the modulus $q = 2m+1$; the smallest such sets are pairs. For q odd let $N = \{v \leq 2m : \gcd(v, q) > 1\}$ be the non-units in the window. The doubling graph on $[1, 2m]$ has edges $\{v, 2v\}$ with $2v \leq 2m$; it lives on the integers of the window, not on residues modulo q.

Proposition 16 (Chain parity)**.** *Let* $q = 2m+1 \geq 3$ *be odd,* $N = \{v \in [1, 2m] : \gcd(v, q) > 1\}$*, and let* Γ *be the graph on* N *with edges* $\{v, 2v\}$*,* $2v \leq 2m$*. If* q *is composite and* $q \neq 9$*, then* $s = q - 2p$*, with* p *the least prime factor of* q*, is an isolated vertex of* Γ*; in particular* Γ *has no perfect matching. If* q *is prime then* $N = \emptyset$*, and if* $q = 9$ *then* $N = \{3, 6\}$ *is a single edge. Hence every component of* Γ *has an even number of vertices if and only if* q *is prime or* $q = 9$*.*

Proof. Let q be composite and p its least prime factor. Then $q = pr$ with $r \geq p \geq 3$, so $q \geq 3p$ and $q \geq p^2$. Put $s = q - 2p$. Then s is odd, $p \leq s \leq q - 6 \leq 2m$, and $\gcd(s, q) = \gcd(2p, q) = p > 1$, so $s \in N$. Being odd, s has no neighbour $s/2$. Its other possible neighbour $2s$ lies in $[1, 2m]$ if and only if $2s \leq q - 1$, that is $q \leq 4p - 1$. In that case $p^2 \leq q \leq 4p - 1$ forces $p \leq 3$, hence $p = 3$ and $q \leq 11$ is an odd composite multiple of 3, so $q = 9$. Thus for $q \neq 9$ the vertex s is isolated, a component with one vertex. If q is prime no $v \in [1, 2m]$ has $\gcd(v, q) > 1$. If $q = 9$, $N = \{3, 6\}$ and Γ is the edge $\{3, 6\}$. An isolated vertex is a component of odd size and admits no perfect matching. □

Label: PROVED. Idea: for composite $q > 9$ *the non-unit* $q - 2p$ *is odd, so it has no halving partner, and its double exceeds the window, so it has no doubling partner; it is an isolated vertex of the doubling graph on the non-units of the window. Zoom map: L1.1* $q - 2p$ *is a non-unit (*$\gcd = p$*); L1.2 it is odd; L1.3 its double exceeds* $2m$ *unless* $q = 9$*. Independent check: brute-force chain lengths (the components of* Γ*) for odd* $q \leq 20001$*, no mismatch (Section 9). Not formalized.*

Remark 17. The companion covering paper classifies every bad pair with non-unit endpoints as a doubling pair or a pair with $a + b \in \{2m + 1, 2m + 2\}$ (Lean-verified for the dichotomy; carried over), and pairs with $a + b = 2m + 1$ and both endpoints non-units do not occur. For $q \in \{25, 27, 33, 35, 49\}$ no non-unit bad pair with $a + b = 2m + 2$ was found in earlier computations, which are not regenerated here (HEURISTIC). If the same holds for all composite q, Proposition 16 implies that a partition of the window into bad pairs cannot cover the non-units, so parts of size at least three are forced; this implication is conditional and carries the label HEURISTIC.

Obstruction for doubling methods. A pairing of the window by $\{j, 2j\}$ treats the integers as a union of doubling chains, and a component with an odd number of vertices leaves one vertex unmatched. For a composite modulus $q > 9$ the vertex $q - 2p$ is an odd non-unit whose double exceeds the window, so it is an isolated vertex of the doubling graph on the non-units; this is the whole mechanism, and it does not depend on any property of the speeds. Passing to the cyclic group $\mathbb{Z}/q\mathbb{Z}$, where doubling is a permutation, does not repair it, since the bad sets of the covering formulation are multiples of j at modulus j and do not realize the wrap-around partner (earlier computations, not regenerated; HEURISTIC).

7 Covering radius and the zero phase

For phases $s \in \mathbb{R}^m$ put $\mathrm{gap}_{v,s}(t) = \min_i \|s_i + v_i t\|$. Henze and Malikiosis identify $\mathrm{ML}(v)$ with $\frac{1}{2} - \frac{1}{2}\kappa(Z(v))$ and the shifted supremum $\sup_t \mathrm{gap}_{v,s}(t)$, minimized over s, with $\frac{1}{2} - \frac{1}{2}\mu(Z(v))$, where κ is the first minimum and μ the covering radius of the lonely runner zonotope $Z(v)$ [4, Prop. 1.7]. Blanco, Criado and Santos show by exact computation that the shifted infimum for $v = (1, \ldots, n)$ is below $1/(n + 1)$ for $5 \le n \le 17$ speeds, for instance $15/94 < 1/6$ for $(1, 2, 3, 4, 5)$ [4, Prop. 5.11, 5.14, Thm. 1.8]; Poliakova shows that the defect grows linearly and is positive for $n \ge 95$ [12, Thm. 1, 2].

Proposition 18. *There is no $t \in \mathbb{R}$ with $\|t\| \ge \frac{1}{7}$, $\|2t - \frac{1}{2}\| \ge \frac{1}{7}$, $\|3t - \frac{1}{2}\| \ge \frac{1}{7}$, $\|4t - \frac{1}{2}\| \ge \frac{1}{7}$, $\|6t - \frac{1}{2}\| \ge \frac{1}{7}$ and $\|9t\| \ge \frac{1}{7}$. That is, for speeds $(1, 2, 3, 4, 6, 9)$ and phases $(0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0)$ the shifted lonely runner bound for seven runners fails.*

Label: VERIFIED, carried over (`halfShift_not_lonely`). Computed value (HEURISTIC, exact): the supremum of the shifted gap is $3/22$, attained at $t = 7/22$, while the zero-phase value for the same speeds is ML $= 1/5$ *(`code/obstructions/shift_grid.py`, exact rational scan of all $t = p/q$, $q \le 400$, and independent recomputation of* ML*).*

Remark 19. By the dictionary above, a bound $\mu(Z(v)) \le (n-1)/(n+1)$ for all v with n speeds is false for $5 \le n \le 17$ (equivalent to the failure of the shifted bound, cited computation, label HEURISTIC as a computation by others, reproduced for one instance by the exact scan above). The covering-radius route of [10], whose Proposition 2.7 bounds $\mu(C) \le \lambda_1(C - C) + \mu(\pi(C), \pi(\mathbb{Z}^d))$ for a convex body C, can therefore be used for the conjecture only through κ, never as a bound at the target value for μ. [10, Lemma 2.5] (if $|C \cap \mathbb{Z}^d| > \ell^d$ then $\lambda_1(C - C) \le 1/\ell$) is a statement on an arbitrary convex body; its conclusion is unchanged by translating C, and the zero phase enters only through the lattice-point count [10, Cor. 2.3].

Obstruction for covering-radius methods. Any argument whose steps remain valid when the zero phase is replaced by a phase vector in $\{0, \frac{1}{2}\}^m$ proves a false statement for $v = (1, 2, 3, 4, 6, 9)$, so a proof of the conjecture must use the zero phase, that is the common starting point of all runners. The covering radius is a translation-invariant quantity and is the right invariant for the shifted problem only; the target quantity is the first minimum. The mechanism is the failure of the shifted statement, proved above for one explicit instance.

8 Descent and topological obstructions

This section records two classes of arguments that do not reach the floor. The first is an induction on the number of speeds that passes from a set to its leave-one-out subsets through an inequality between values of ML. The second is a topological necessary condition for the straight line $t \mapsto tv$ to avoid the cube $[\delta, 1-\delta]^m$ in the torus. In this section V is a set of m distinct positive integers, identified with the vector of its elements, and $V \setminus v$ is V without the element v. For $m \geq 2$ put

$$\mathrm{loo}(V) = \min_{v \in V} \mathrm{ML}(V \setminus v).$$

A *tight leave-one-out* of V is an element v with $\mathrm{ML}(V \setminus v) = 1/m$, the value of the floor for $m-1$ speeds. For $k \geq 1$ put

$$\sigma(k) = \inf\{\mathrm{ML}(W) : \ W \text{ a set of } k \text{ distinct positive integers, } \mathrm{ML}(W) > 1/(k+1)\}.$$

No value of ML at k speeds lies strictly between $1/(k+1)$ and $\sigma(k)$.

8.1 Deletion and dilation

Lemma 20. *Let V be a finite multiset of positive integers with at least two elements and let $c \geq 1$ be an integer. (a) For every element v of V, $\mathrm{ML}(V \setminus v) \geq \mathrm{ML}(V)$. (b) $\mathrm{ML}(cV) = \mathrm{ML}(V)$.*

Proof. (a) For every t, $\mathrm{gap}_{V \setminus v}(t)$ is a minimum over a subset of the terms of $\mathrm{gap}_V(t)$, so $\mathrm{gap}_{V \setminus v}(t) \geq \mathrm{gap}_V(t)$; take the maximum over t. (b) $\mathrm{gap}_{cV}(t) = \mathrm{gap}_V(ct)$, and $t \mapsto ct$ maps $\mathbb{R}$ onto $\mathbb{R}$, so the two functions have the same range. □

Label: PROVED. Idea: deleting a speed removes a constraint, and dilating all speeds reparametrizes time. Zoom map: L1.1 pointwise inequality of the gaps; L1.2 surjectivity of $t \mapsto ct$. Not formalized (a Lean statement of (b) for tuples exists in the source tree as `ML_smul`, see `LEAN_MAP.md`; (a) is absent). Elementary; the corresponding deletion statement for the shifted problem is [17, Lemma 7.14], a citation that was not re-checked in this revision.

Remark 21. Lemma 20 bounds ML of a subset from below by ML of the whole set. It does not give $\mathrm{ML}(V \setminus v) \geq 1/m$, which is the floor for $m-1$ speeds. In particular it does not show that a subset of a counterexample with m speeds is not a counterexample with $m-1$ speeds; that is the conjecture at $m-1$ speeds applied to the subset.

8.2 Leave-one-out descent

For $m \geq 2$ a *descent function at m* is a function $F : (0, \frac{1}{2}] \to \mathbb{R}$ such that

$$\mathrm{loo}(V) \leq F(\mathrm{ML}(V)) \qquad \text{for every set } V \text{ of } m \text{ distinct positive integers.} \tag{2}$$

It is *sub-floor* if $F(x) < \sigma(m-1)$ for all $0 < x < 1/(m+1)$. The point of the definition is the following. Suppose the floor holds for $m-1$ speeds and F is a sub-floor descent function at m. If $\mathrm{ML}(V) < 1/(m+1)$, then some v has $\mathrm{ML}(V \setminus v) < \sigma(m-1)$, hence $\mathrm{ML}(V \setminus v) = 1/m$ because $\mathrm{ML}(V \setminus v) \geq 1/m$ and no value lies in $(1/m, \sigma(m-1))$. So a sub-floor descent function would send every counterexample with m speeds to a tight set with $m-1$ speeds.

Lemma 22 (Descent lemma)**.** *If F is a descent function at m, then $F(1/(m+1)) \geq \mathrm{loo}(V)$ for every tight V with m speeds.*

Proof. Apply (2) to V, for which $\mathrm{ML}(V) = 1/(m+1)$. □

Label: PROVED. Idea: a descent function is also required to be correct on the tight sets, and there it must be at least as large as their leave-one-out values. Zoom map: L1.1 the defining inequality at V. Not formalized. The quantifier is over all sets of m distinct speeds, tight ones included; it is not a statement about all tight sets, only about each one.

Observation 23 (Exact instances). (a) $V = (1, 4, 5, 6, 7, 11, 13)$ and $(1, 2, 3, 4, 5, 7, 12)$ satisfy $\mathrm{ML}(V) = 1/8$ and $\mathrm{loo}(V) = 2/13$ (removing 13, respectively 7). $W = (1, 5, 6, 11, 16, 17)$ has six speeds and $\mathrm{ML}(W) = 5/33$, and $1/7 < 5/33 < 2/13$; hence $\sigma(6) \leq 5/33 < 2/13$.
(b) For $3 \leq m \leq 100$ and for $m \in \{121, 143, 151, 181, 211, 241\}$, the set $[m-2] \cup \{2(m-1)\}$ has $m-1$ speeds and $\mathrm{ML} = 2/(2m-1) > 1/m$; hence $\sigma(m-1) \leq 2/(2m-1)$.
(c) For each $m \in M = \{7, 31, 61, 73, 91, 121, 143, 151, 181, 211, 241\}$ there is a tight V_m with m speeds and $\mathrm{loo}(V_m) > \sigma(m-1)$, listed in Table 2: V_7 as in (a) with $\sigma(6) \leq 5/33$, and for $m \neq 7$ a set $[m] \setminus \{r\} \cup \{w\}$ with $\mathrm{loo}(V_m) > 2/(2m-1) \geq \sigma(m-1)$.

Label: HEURISTIC. Exact rational computation with a routine that enumerates every numerator k of the candidate times $k/(v_i + v_j)$ (script `code/obstructions/descent_check.py`, modes `controls`, `instances`, `strict`; outputs in `data/obstructions`); not formalized; the case $m = 7$ can be upgraded to VERIFIED by a kernel computation. Ground: in Table 2, $\mathrm{ML}(V_m) = 1/(m+1)$ and $\mathrm{loo}(V_m)$ over all m removals are recomputed for $m \leq 73$; $\mathrm{ML}(V_m)$ and one removal value for $m \leq 151$; the remaining entries are taken from a table of an exact job whose source is not archived (generator not archived). Termination: RESOLVED (finite evaluation). Earlier variants of the routine omitted numerators not coprime to the denominator and are not used.

Proposition 24 (Descent discontinuity)**.** *Let $m \in M$ and let F be a sub-floor descent function at m. Then*

$$\sup_{0<x<1/(m+1)} F(x) \leq \sigma(m-1) < \mathrm{loo}(V_m) \leq F\Big(\tfrac{1}{m+1}\Big).$$

In particular F is not left-continuous at $1/(m+1)$, and its jump there is at least $\mathrm{loo}(V_m) - \sigma(m-1) > 0$; at $m = 7$ this is at least $1/429$.

Proof. The first inequality is the definition of sub-floor, the second is Observation 23(c), the third is Lemma 22 for V_m. □

Label: HEURISTIC (effective label). The deduction from Lemma 22 and Observation 23(c) is PROVED; each listed m rests on the exact computations of the observation, which are not formalized. Zoom map: L1.1 Lemma 22; L1.2 a tight set with loo *above $\sigma(m-1)$ (Observation 23); L1.3 comparison of the two sides. The proposition is not asserted at other m; at $m = 4, 5, 13, 19$ the sets found have $\mathrm{loo}(V) = 2/(2m-1)$ and the comparison with $\sigma(m-1)$ was not decided here.*

m	V_m	$\mathrm{loo}(V_m)$	$2/(2m-1)$	upper bound for $\sigma(m-1)$	recomputed
7	$(1,4,5,6,7,11,13)$	2/13	2/13	5/33	all removals
31	$[31]\setminus\{30\}\cup\{90\}$	3/91	2/61	2/61	all removals
61	$[61]\setminus\{60\}\cup\{180\}$	3/181	2/121	2/121	all removals
73	$[73]\setminus\{70\}\cup\{140\}$	1/72	2/145	2/145	all removals
91	$[91]\setminus\{90\}\cup\{270\}$	3/271	2/181	2/181	V, one removal
121	$[121]\setminus\{120\}\cup\{360\}$	3/361	2/241	2/241	V, one removal
143	$[143]\setminus\{140\}\cup\{280\}$	1/142	2/285	2/285	V, one removal
151	$[151]\setminus\{150\}\cup\{450\}$	3/451	2/301	2/301	V, one removal
181	$[181]\setminus\{180\}\cup\{540\}$	3/541	2/361	2/361	table only
211	$[211]\setminus\{210\}\cup\{630\}$	3/631	2/421	2/421	table only
241	$[241]\setminus\{240\}\cup\{720\}$	3/721	2/481	2/481	table only

Table 2: Tight sets V_m with $\mathrm{loo}(V_m) > \sigma(m-1)$. The last column states what was recomputed in this archive. At $m = 73$ the set with the two entries $70, 72$ replaced by $140, 144$ has the same loo; at $m = 211$ the replacements $210 \to 840$ and $210 \to 1050$ give 4/841 and 5/1051. The last column of upper bounds is Observation 23(b), except at $m = 7$.

Class excluded and class not excluded. The proposition concerns functions of $\mathrm{ML}(V)$ alone that satisfy (2) for all sets of m distinct speeds, tight sets included, that stay below $\sigma(m-1)$ under the floor and that are left-continuous at the floor, and it concerns deletion of one speed. It does not exclude: inequalities with structural hypotheses on V (leave-one-out primitivity, divisor-completeness, a given relation); inequalities whose right-hand side uses other invariants of V; inequalities with a threshold above $\sigma(m-1)$ followed by a classification of the sets that are near the floor; deletion of two or more speeds (the first two near-tight sets of Observation 27 contain tight subsets of several smaller sizes); functions with a jump at the floor, which satisfy (2) vacuously below the floor if the floor holds for m speeds and carry no information about counterexamples. The left side of (2) is itself a value of ML of subsets; what is excluded is a function of $\mathrm{ML}(V)$ alone with the stated properties.

Conceptual abstract. The move: to run an induction on the number of speeds, ask for an inequality that sends a hypothetical counterexample with m speeds to a subset with $m-1$ speeds forced to be tight, namely $\min_v \mathrm{ML}(V\setminus v)$ is below the least value above the floor that $m-1$ speeds can take, whenever $\mathrm{ML}(V)$ is below the floor, with a bound that depends on V only through $\mathrm{ML}(V)$ and is continuous at the floor. The exchange: an existence statement becomes an inequality between two numbers. The mechanism that defeats it: some tight sets (Goddyn–Wong sets with one fast speed, and the $m = 7$ set $(1,4,5,6,7,11,13)$) have every leave-one-out value strictly above that value, so the bound must jump exactly at the floor. Left open: inequalities that look at lonely times, or at structure, or delete two speeds.

8.3 Leave-one-out values of the known tight sets

Observation 25 (Seventy tight sets). Consider the following 70 tight sets: the 66 sets $[n-1]$ with one entry r replaced by μr under the GCD condition for $3 \le n \le 260$ (regenerated by `descent_check.py gw`), the set $[73]\setminus\{70,72\}\cup\{140,144\}$ with two replaced entries, and the sporadic sets $(1,3,4,7)$, $(1,3,4,5,9)$, $(1,4,5,6,7,11,13)$; call the first 67 the Goddyn–Wong type sets. Then:

Class	Number	Values of m or composition
with a tight leave-one-out ($\mathrm{loo} = 1/m$)	11	$32, 62, 92, 122, 152, 182, 212$ (twice), $213, 214, 242$
without a tight leave-one-out	59	$= 56$ Goddyn–Wong type $+ 3$ sporadic
$\mathrm{loo}(V) = 2/(2m-1)$	46	$= 43$ Goddyn–Wong type $+ 3$ sporadic
$\mathrm{loo}(V) > 2/(2m-1)$	13	$31, 61, 73$ (twice), $91, 121, 143, 151, 181, 211$ (three), 241
$\mathrm{loo}(V) < 2/(2m-1)$	0	

The number 43 counts the Goddyn–Wong type sets only; 46 includes the three sporadic sets. The 13 sets above $2/(2m-1)$ are all of Goddyn–Wong type.

Label: HEURISTIC. Exact computation (`descent_check.py table`; output `data/obstructions/descent_table.out`). Ground: the 70 *sets; the leave-one-out values are recomputed here for $m \le 73$ (21 sets) and taken from a table of an exact job whose source is not archived for the other* 49 *(generator not archived); tightness is recomputed for the 21 sets with $m \le 73$ and for the sets of Table 2 up to $m = 151$, and otherwise rests on [8, Thm. 3.1] (single replacement under the GCD condition) and on [17] (the double replacement). The list is not a classification: it is the list of known tight sets with a bounded number of replaced entries. Termination: RESOLVED.*

Conjecture 26 ((C1))**.** *Let $m \ge 2$ and let V be a tight set of m distinct positive integers with $\gcd(V) = 1$ and no tight leave-one-out. Then $\mathrm{loo}(V) \ge 2/(2m-1)$.*

Label: CONJECTURE. Provenance: observation on the table above. Evidence: the 59 *sets without a tight leave-one-out in Observation 25 satisfy it, in three structurally different classes (single acceleration, double acceleration, sporadic), and the exhaustive boxes of Section 9 ($m = 3, \dots, 8$) contain no tight set other than $[m]$ and sets of Observation 25. Consequence: any descent function at m satisfies $F(1/(m+1)) \ge 2/(2m-1)$ for every m at which such a set exists. The universal statement "every tight set has $\mathrm{loo}(V) \ge 2/(2m-1)$" is false: the* 11 *sets with a tight leave-one-out and the sets $[m]$ have $\mathrm{loo} = 1/m$. Proposition 24 does not use (C1).*

Observation 27 (Near-tight sets without a tight leave-one-out). The statement "if $\mathrm{ML}(V) < 1/m$ then $\mathrm{ML}(V \setminus v) = 1/m$ for some v" is false. Each of the following has $1/(m+1) < \mathrm{ML}(V) < 1/m$ and no tight leave-one-out:

m	V	$\mathrm{ML}(V)$	$\mathrm{loo}(V)$ (removed)	tight subsets of size $2, \dots, m-1$
5	$(1, 2, 3, 5, 8)$	2/11	2/9 (5)	$(1, 2)$, $(1, 2, 3)$; 2 of 25
8	$(1, 2, 3, 4, 5, 7, 18, 40)$	5/41	3/23 (40)	$(1, 2)$, $(2, 4)$, $(1, 2, 3)$, $(1, 2, 3, 4)$, $(1, 3, 4, 7)$, $(1, 2, 3, 4, 5)$; 6 of 246
10	$(1, 5, 6, 7, 8, 11, 13, 19, 20, 27)$	3/31	2/17 (20)	none; 0 of 1012

The first two sets have tight subsets with fewer elements, so for them the corresponding statement for subsets of any size $2, \dots, m-1$ is not refuted. The third set has no tight subset of any size $2, \dots, 9$.

Label: FALSE (the quoted statement), witnessed by the three exact computations above (`descent_check.py instances`; output `data/obstructions/descent_instances.out`); the computations are not formalized. The same statement with "leave-one-out" replaced by "subset of size $2, \dots, m-1$" is refuted by the third set only.

Remark 28. A descent carried by lonely times of the subsets, rather than by values of ML, is outside Proposition 24. Whether such a descent exists is not known. Label: HEURISTIC, as this delimits a method.

8.4 Topology of the bad region

Let $\mathbb{T}^m = \mathbb{R}^m/\mathbb{Z}^m$ and $q : \mathbb{R}^m \to \mathbb{T}^m$ the quotient map. For $0 < \delta \le \frac{1}{2}$ put

$$U_\delta = \{x \in \mathbb{T}^m : \ \|x_i\| < \delta \text{ for some } i\}, \qquad C_\delta = \mathbb{T}^m \setminus U_\delta = q([\delta, 1-\delta]^m).$$

U_δ is the union of the m open slabs around the coordinate subtori; C_δ is one closed cube (a point if $\delta = \frac{1}{2}$) in the open fundamental cube $q((0,1)^m)$. For $v \in \mathbb{Z}^m$ let $\gamma_v : \mathbb{R}/\mathbb{Z} \to \mathbb{T}^m$, $\gamma_v(t) = q(tv)$. Then $t \in G_\delta$ if and only if $\gamma_v(t) \in C_\delta$, so $\mathrm{ML}(v) \ge \delta$ if and only if γ_v meets C_δ.

Proposition 29 (Bad region)**.** *Let $0 < \delta \le \frac{1}{2}$ and let $i : U_\delta \to \mathbb{T}^m$ be the inclusion.*
(a) For $m \ge 2$, $i_* : H_1(U_\delta; \mathbb{Z}) \to H_1(\mathbb{T}^m; \mathbb{Z}) = \mathbb{Z}^m$ is an isomorphism.
(b) For $m \ge 3$, $i_* : \pi_1(U_\delta, x_0) \to \pi_1(\mathbb{T}^m, x_0)$ is an isomorphism for every $x_0 \in U_\delta$. For $m = 2$ and every $x_0 \in U_\delta$ it is surjective and not injective.
(c) For $m = 1$, U_δ is an open arc and i_* on H_1 is not an isomorphism.

Proof. Write $c = (\frac{1}{2}, \dots, \frac{1}{2})$, $\rho(x) = \|x - c\|_\infty$ for $x \in [0,1]^m$, and identify $(0,1)^m$ with its image under q. Then $C_\delta = \{\rho \le \frac{1}{2} - \delta\}$. Put $B = \{x \in (0,1)^m : \rho(x) < \frac{1}{2} - \frac{\delta}{2}\}$, an open cube. Since $\delta > 0$, $C_\delta \subset B$, so $\mathbb{T}^m = U_\delta \cup B$, and $U_\delta \cap B = \{\frac{1}{2} - \delta < \rho < \frac{1}{2} - \frac{\delta}{2}\}$. Since $\rho > \frac{1}{2} - \delta \ge 0$ on $U_\delta \cap B$, $c \notin U_\delta \cap B$, and $x \mapsto \big((x-c)/\rho(x), \rho(x)\big)$ is a homeomorphism of $U_\delta \cap B$ onto $S \times (\frac{1}{2} - \delta, \frac{1}{2} - \frac{\delta}{2})$, where $S = \{y : \|y\|_\infty = 1\} \cong S^{m-1}$. The set U_δ is path connected: each slab is connected and contains 0. The set B is contractible.

(a) and (b), $m \ge 3$. S^{m-1} is connected and has $H_1 = 0$ and $\pi_1 = 1$. The Mayer–Vietoris sequence of the open cover $\{U_\delta, B\}$ contains

$$H_1(U_\delta \cap B) \to H_1(U_\delta) \oplus H_1(B) \to H_1(\mathbb{T}^m) \to \widetilde{H}_0(U_\delta \cap B),$$

in which $H_1(U_\delta \cap B) = H_1(S^{m-1}) = 0$, $\widetilde{H}_0(U_\delta \cap B) = 0$ and $H_1(B) = 0$; the middle map is i_* on the first summand. Hence i_* is an isomorphism on H_1. By the van Kampen theorem for the open path-connected sets U_δ, B with path-connected intersection, $\pi_1(\mathbb{T}^m)$ is the amalgam of $\pi_1(U_\delta)$ and $\pi_1(B) = 1$ over $\pi_1(U_\delta \cap B) = 1$, that is $\pi_1(U_\delta)$, the identification being induced by i. Path connectedness of U_δ makes the statement independent of the base point.

(a) and (b), $m = 2$. Let $\lambda(x) = 2\rho(x) \in (0,1]$ on $[0,1]^2 \setminus \{c\}$ and $h_s(x) = c + (1 - s + s/\lambda(x))(x - c)$, $s \in [0,1]$. Then $\lambda(h_s(x)) = (1-s)\lambda(x) + s \ge \lambda(x)$, h_s is the identity on $\partial[0,1]^2$ (where $\lambda = 1$), and h_1 maps into $\partial[0,1]^2$. Since $U_\delta = q(\{\lambda > 1 - 2\delta\})$ and q identifies only points of $\partial[0,1]^2$, h_s induces a strong deformation retraction of U_δ onto $Z = q(\partial[0,1]^2) = \{x_1 = 0\} \cup \{x_2 = 0\} \subset U_\delta$. The set Z is a wedge of two circles $a_1 = \{x_2 = 0\}$, $a_2 = \{x_1 = 0\}$; $H_1(Z) = \mathbb{Z}^2$ with basis $[a_1], [a_2]$, and $\pi_1(Z)$ is free on a_1, a_2. The loops a_1, a_2 represent the standard generators of $H_1(\mathbb{T}^2) = \pi_1(\mathbb{T}^2) = \mathbb{Z}^2$. So i_* is an isomorphism on H_1. On $\pi_1(U_\delta, 0) = F(a_1, a_2)$ it is the abelianization $F_2 \to \mathbb{Z}^2$, hence surjective, and not injective because the commutator of a_1, a_2 is nontrivial in F_2; for another base point x_0 use a path in U_δ from 0 to x_0.

(c) $\mathbb{T}^1 = \mathbb{R}/\mathbb{Z}$ and $U_\delta = q((-\delta, \delta))$ is an arc, $H_1(U_\delta) = 0 \ne \mathbb{Z} = H_1(\mathbb{T}^1)$. □

Label: PROVED. Hypotheses used: $\delta > 0$ for $C_\delta \subset B$; $\frac{1}{2} - \delta \geq 0$ so that the shell $U_\delta \cap B$ is bounded away from the centre c ($\delta < \frac{1}{2}$ is not needed: the statement and proof hold for $0 < \delta \leq \frac{1}{2}$, and for $\delta > \frac{1}{2}$ the set U_δ is the whole torus); $m \geq 2$ for the connectedness of the shell; $m \geq 3$ for its simple connectivity Prior art: removing a ball from a manifold of dimension at least 3 does not change π_1, and from a torus of dimension at least 2 does not change H_1; this is standard, and no novelty is claimed for it or for its application here; no literature search was made for the lonely runner formulation.

Conceptual abstract. The central idea is to ask what the inclusion of the bad region into the torus can detect about a speed vector. The bad region, where some runner is within δ of the origin, is the torus minus one cube; the sphere around the cube is connected, and simply connected from dimension three, so the inclusion is an isomorphism on first homology in dimension at least two and on the fundamental group in dimension at least three. Hence the curve traced by a speed vector is homotopic into the bad region, as lifting also shows. For curves symmetric under negation the one remaining obstruction is the midpoint, a fixed point in the cube exactly when all speeds are odd; otherwise half the curve is a path between fixed points, pushed in by lifting. So such conditions detect only the all-odd vectors, where the maximum is already $\frac{1}{2}$.

Zoom map for Propositions 29 and 31.

L1.1 The bad region is the torus minus one closed cube C_δ (definition of U_δ; $C_\delta \subset B$, $U_\delta \cap B \cong S^{m-1} \times$ interval).

L1.2 The inclusion $U_\delta \to \mathbb{T}^m$ is an isomorphism on H_1 for $m \geq 2$ and on π_1 for $m \geq 3$, and is onto on π_1 for $m = 2$ (Mayer–Vietoris for $m \geq 3$; van Kampen for $m \geq 3$; deformation retraction onto the wedge of two circles for $m = 2$).

L1.3 The only obstruction to an equivariant push of γ_v into U_δ is the position of $\gamma_v(\frac{1}{2}) = v/2 \bmod 1$, which equals p if and only if all v_i are odd (equivariant loops are paths between fixed points; path lifting in $\mathbb{R}^m$, one coordinate at a time, with another coordinate at an integer).

Corollary 30. *Let $m \geq 2$, $v \in \mathbb{Z}^m$ and $0 < \delta \leq \frac{1}{2}$. There is a loop in U_δ based at 0 that is homotopic to γ_v in $\mathbb{T}^m$ rel the base point 0; in particular $[\gamma_v] \in \mathrm{im}(i_* : H_1(U_\delta) \to H_1(\mathbb{T}^m))$.*

Proof. γ_v is the image under q of the segment from 0 to v. Let σ be the path in $\mathbb{R}^m$ that first moves x_1 from 0 to v_1, with $x_2 = 0$, and then moves x_k from 0 to v_k for $k = 2, \ldots, m$ in turn, with $x_1 = v_1 \in \mathbb{Z}$. Every point of σ has a coordinate that is an integer, so $q \circ \sigma$ lies in U_δ; the first leg uses $m \geq 2$. The straight-line homotopy in $\mathbb{R}^m$ from σ to the segment fixes the endpoints 0 and v, and its image under q is a homotopy rel 0 in $\mathbb{T}^m$ from $q \circ \sigma$ to γ_v. □

Label: PROVED. The proof does not use Proposition 29; it uses $\delta > 0$ and $m \geq 2$ only (for $m = 1$ and $v = 2$ the statement fails, as U_δ is an arc). Positivity and primitivity of v are not used and not claimed.

An *equivariant loop* is a loop $\gamma : \mathbb{R}/\mathbb{Z} \to \mathbb{T}^m$ with $\gamma(-t) = -\gamma(t)$; an *equivariant homotopy* is a homotopy through equivariant loops. The loop γ_v is equivariant. The fixed points of $x \mapsto -x$ on $\mathbb{T}^m$ are the 2^m points of $(\frac{1}{2}\mathbb{Z}/\mathbb{Z})^m$; exactly one of them, $p = (\frac{1}{2}, \ldots, \frac{1}{2})$, lies in C_δ.

Proposition 31 (Parity)**.** *Let $m \geq 2$, $0 < \delta \leq \frac{1}{2}$ and $v \in \mathbb{Z}^m$. The loop γ_v is equivariantly homotopic to an equivariant loop with image in U_δ if and only if not all v_i are odd.*

Proof. Let F be the set of fixed points of $x \mapsto -x$, a finite set. An equivariant loop γ has $\gamma(0), \gamma(\frac{1}{2}) \in F$, because $t = 0$ and $t = \frac{1}{2}$ are fixed by $t \mapsto -t$, so equivariance gives $\gamma(t) = -\gamma(t)$ there. During an equivariant homotopy these two points move continuously in the finite set F, so they are constant.

If all v_i are odd, then $\gamma_v(\frac{1}{2}) = p \notin U_\delta$, so every equivariantly homotopic loop passes through p.

Suppose not all v_i are odd, and choose i with v_i even. Then $b = \gamma_v(\frac{1}{2}) = v/2 \bmod 1$ has i-th coordinate 0, so $b \in F \setminus \{p\}$ and $b \in U_\delta$. Let $\beta = \gamma_v|_{[0,1/2]}$, the image under q of the segment from 0 to $v/2$. Let σ be the path in $\mathbb{R}^m$ that first moves x_i from 0 to $v_i/2 \in \mathbb{Z}$, with another coordinate $x_j = 0$ (this uses $m \geq 2$), and then moves each other coordinate x_k from 0 to $v_k/2$ in turn, with $x_i \in \mathbb{Z}$. Every point of σ has a coordinate that is an integer, so $\beta' = q \circ \sigma$ lies in U_δ, and the straight-line homotopy in $\mathbb{R}^m$ from σ to the segment, composed with q, is a homotopy rel endpoints in $\mathbb{T}^m$ from β to β'; let $H : [0, \frac{1}{2}] \times [0, 1] \to \mathbb{T}^m$ be it, $H(\cdot, 0) = \beta$, $H(\cdot, 1) = \beta'$. Define $\gamma_s(t) = H(t, s)$ for $t \in [0, \frac{1}{2}]$ and $\gamma_s(t) = -H(-t, s)$ for $t \in [-\frac{1}{2}, 0]$. The two definitions agree at $t = 0$ ($H(0, s) = 0 = -0$) and at $t = \pm\frac{1}{2}$ ($-b = b$), so each γ_s is an equivariant loop, $\gamma_0 = \gamma_v$, and the image of γ_1 is $\beta'([0, \frac{1}{2}]) \cup -\beta'([0, \frac{1}{2}]) \subset U_\delta$, since $U_\delta = -U_\delta$. □

Label: PROVED. Idea: an equivariant loop is a path between two fixed points, and fixed points cannot move; a path between two fixed points in $\mathbb{T}^m$ is lifted to $\mathbb{R}^m$, where it can be replaced by a path through the slabs that is homotopic to it rel endpoints. Zoom map: L1.1 equivariant loops correspond to paths between fixed points; L1.2 the endpoint $\gamma_v(\frac{1}{2})$ is p exactly for all v_i odd; L1.3 path lifting in $\mathbb{R}^m$. The proof does not use Proposition 29 or Corollary 30. Hypotheses used: $m \geq 2$ in the first leg of σ (fails for $m = 1$, $v = 2$); $\delta \leq \frac{1}{2}$ only for $p \notin U_\delta$ and for p being the only fixed point in C_δ; positivity, distinctness, primitivity and gcd of v are not used and not claimed If all v_i are odd then $\mathrm{ML}(v) = \frac{1}{2}$ (the parity trick). The same argument for phase shifts in $\{0, \frac{1}{2}\}^m$ is not stated here.

Remark 32 (What the topology excludes). Corollary 30 and Proposition 31 say that a necessary condition for $\gamma_v \subset U_\delta$ obtained from the inclusion $U_\delta \to \mathbb{T}^m$ (membership of $[\gamma_v]$ in the image of H_1 or π_1, or the equivariant version) holds for every v with not all v_i odd, so such a condition never yields a contradiction with $\mathrm{ML}(v) < \delta$. The statement is about arguments of this form. It is not the statement that no function of the class $[\gamma_v] = v$ distinguishes $\mathrm{ML}(v) \geq \delta$ from $\mathrm{ML}(v) < \delta$: ML is a function of v. What is absent from the topological data is the straight-line parametrization of γ_v. For v with all v_i odd the equivariant condition fails, and $\mathrm{ML}(v) = \frac{1}{2} \geq \delta$, so no contradiction is available there either. No statement is made here about degrees, Euler classes or other invariants. Label: HEURISTIC, as this delimits a method; the propositions it rests on are PROVED.

Remark 33 (Parity and the value $\frac{1}{2}$). For $v \in \mathbb{Z}^m_{>0}$, $\mathrm{ML}(v) = \frac{1}{2}$ if and only if all v_i have the same 2-adic valuation. Proof. If $v_i = 2^k u_i$ with u_i odd for all i, then at $t = 2^{-k-1}$ every $v_i t = u_i/2$ has $\|v_i t\| = \frac{1}{2}$. Conversely, if $\mathrm{ML}(v) = \frac{1}{2}$, a time t with $\|v_i t\| = \frac{1}{2}$ for all i satisfies $v_i t \in \frac{1}{2} + \mathbb{Z}$, so $t = r/s$ in lowest terms and $2v_i r = (2a_i + 1)s$ with $a_i \in \mathbb{Z}$; comparing 2-adic valuations, $1 + \nu_2(v_i) + \nu_2(r) = \nu_2(s)$ for every i, so $\nu_2(v_i)$ does not depend on i. The parity condition of Proposition 31 is not invariant under $v \mapsto cv$, while ML is (Lemma 20(b)); there is no conflict, since Proposition 31 concerns one necessary condition. Example: $\mathrm{ML}(2, 6) = \frac{1}{2}$.

Label: PROVED (two lines; checked by the author only; also computed with the standard routine for 200 random vectors, 104 with ML $= \frac{1}{2}$, no mismatch).

9 Evidence and regeneration

All commands are run from the archive root; searches report RESOLVED or UNRESOLVED. Cases counted are independent vectors in distinct boxes.

Item	Regeneration command	Status
ML of the five rows of Table 1, of E_{14} and of $(1,2,3,4,6,9)$	`mlcheck` (Rust) via `code/obstructions/repro_04.sh`; `python3 code/obstructions/mlexact.py` (independent, Python, all numerators, exact integers)	RESOLVED; both programs agree on 14/121, 11/84, 168/1697, 28/191, 30/203, 1/5
Certificate levels, U_1, U_2	`code/obstructions/repro_04.sh`	RESOLVED
Census and boxes	`mdr enumw M H`, `mdr cert 80 FILE` (see `code/obstructions/README.md`)	RESOLVED
Last row of Table 1	`MDR_WX=1 mdr hunt2` (README)	UNRESOLVED
Tight vectors, (T), (T′)	`python3 code/obstructions/tightscan.py`	see below
Minimum of ML on DC vectors (T2)	`python3 code/obstructions/t2box.py`	see below
Goddyn–Wong type vectors	`python3 code/obstructions/gwcheck.py`	RESOLVED
Chain parity	`python3 code/obstructions/chainparity.py 20001`	RESOLVED, $q \leq 20001$
Shifted supremum	`python3 code/obstructions/shift_grid.py`	RESOLVED
Controls of the ML routine; Lemma 20 on random vectors	`descent_check.py controls`, `d0` (in `code/obstructions`)	RESOLVED
Observations 23, 27	`descent_check.py instances`, `strict`	RESOLVED
Observation 25	`descent_check.py gw`, `table`	RESOLVED; 49 of 70 rows from a table, see label

Tight vectors. For each m and bound H, all primitive $v \in [1,H]^m$ with distinct entries such that every $j \in [2,m]$ divides a speed (necessary for tightness by Proposition 6) were enumerated and ML computed exactly, together with all maximizing times.

m	H	vectors enumerated	tight vectors found
3	60	17 432	$(1,2,3)$
4	40	47 078	$(1,2,3,4)$, $(1,3,4,7)$
5	30	63 880	$(1,\dots,5)$, $(1,3,4,5,9)$
6	24	57 111	$(1,\dots,6)$
7	20	23 212	$(1,\dots,7)$, $(1,2,3,4,5,7,12)$, $(1,4,5,6,7,11,13)$
8	16	4 481	$(1,\dots,8)$

In every box, no tight vector has a speed divisible by $m+1$, and every maximizing time of every tight vector lies in $\frac{1}{m+1}\mathbb{Z}$. Each box is RESOLVED; the boxes are small (toy only), the heights for $m \geq 6$ are at most 24, and the boxes for different m are independent instances (different numbers of speeds). The evidence for (T) and (T′) is confined to this ground and says nothing for $m \geq 9$ or beyond these heights; both remain CONJECTURE.

Minimum on divisor-complete vectors. In the same boxes (primitive, DC including $j = m+1$) the minimum of ML is $2/7$ ($m = 3$, $H \leq 60$; 10 946 vectors), $2/9$ ($m = 4$, $H \leq 40$; 27 600), $2/11$ ($m = 5$, $H \leq 30$; 46 316), $2/13$ ($m = 6$, $H \leq 24$; 28 835), $2/15$ ($m = 7$, $H \leq 20$; 12 673), each equal to $2/(2m+1)$, and strictly below $\mathrm{ML}(E_m) = 4/13, 5/21, 6/31, 7/43, 8/57$. Each box is RESOLVED; the boxes are small, and the evidence is HEURISTIC and toy only: no statement is made for $m \geq 8$. The tight and DC counts in neighbouring boxes are not independent.

10 Open statements

- (T) and (T′), Conjecture 12. Evidence: the exhaustive boxes above and every tight vector of Goddyn–Wong type (Corollary 10). Consequence: (T) is equivalent to the statement that no tight primitive vector is DC (Conjecture 12 and Proposition 6); (T′) implies (T) (Proposition 13).
- (T2), Conjecture 15. Evidence: equality $2/(2m+1)$ is the minimum on DC vectors in the boxes for $m = 3, \ldots, 7$. Consequence: strict floor for primitive vectors with $(m+1) \mid v_i$.
- (C1), Conjecture 26. Evidence: the 59 known tight vectors without a tight leave-one-out. Consequence: every descent function at m is at least $2/(2m-1)$ at the floor wherever such a vector exists.

Whether the class of the crux contains a tight vector is open. None of the three statements is used elsewhere in this note.

11 Attribution and prior art

Each item was compared with the sources cited; “known” means the statement or its proof is in the cited place.

Item	Status	Reference location
Thm 2, inequality for weights $F \geq 0$	known	[3, Lemma 2.1] with $\mu = F\,dt$ (closed bad sets, ML $= \delta$); union bound [15, Prop. 1.1]
Thm 2, countability of G_δ; formalization	ours	elementary; the Lean formalization is ours
Cor 3	known in substance	consequence of the same lemma; the vanishing $H \equiv 0$ at tight instances is used in [2, proof of Thm. 2.1]
Prop 5	known in substance	[16, Def. 3], [13] for $\varepsilon = 0$; $\varepsilon \neq 0$ routine
Prop 6	known	[15, Lemma 5.1(i)]; [17, Lemma 2.3]
Prop 7	ours	no statement located in the sources read
Obs 8, Table 1	ours	computation; no comparable table located
Thm 9, Cor 10	partially known	the tightness statements are [8, Thm. 3.1, 2.3], [17, Thm. 1.1, 1.2, 5.1]; the non-divisibility corollary was not located (original proof of [8] not read)
Prop 13	ours	elementary; (T) is not stated in [9], [17], [5], [11] as read. [5, Thm. 4.2] uses the non-divisibility of the four tight instances with $n \leq 6$ as an observation
Conj 12, 15	ours	see above; [9, Cor. 2.3 for E_m] is [9, Thm. 5.1]
Prop 16	ours	elementary; no prior statement located
Prop 18	ours (instance)	the shifted conjecture is false: [4, Thm. 1.8]; the instance $(1, 2, 3, 4, 6, 9)$ with half-phases is a second, independently checked counterexample, not claimed as new in kind
Rem 19	known	[4, Prop. 1.7], [10, Prop. 2.7, Lemma 2.5]
Lem 20	elementary	shifted deletion statement [17, Lemma 7.14] (not re-checked); scaling invariance is folklore
Lem 22, Obs 23, Prop 24, Obs 25, 27, Conj 26	ours (as stated)	no statement on leave-one-out values of tight vectors or on descent inequalities located in [7], [5], [17], [2], [6] as read; no web search; tightness of the vectors is [8, Thm. 3.1], [17]
Prop 29, Cor 30, Prop 31	standard topology	removal of a ball from a torus; no novelty claimed; no literature search for the lonely runner formulation

The qualitative statement that counterexamples and tight vectors lie on finitely many explicit hyperplanes is [2, Thm. 1.1, 2.1] (norm $2m + 3$), derivable with weaker bounds from [7]; the random-speed precursor is [6]. Nothing in this note claims it.

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